

CAMILLE TRAN

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🐙 github.com/Ctt011 🌐 ctt011.github.io/Camille-Tran-Website/

Career Objective

Data Science graduating senior (June 2026) with practical experience across data engineering, machine learning, and data visualization from industry-focused projects and internships. Built end-to-end ML pipelines, real-time streaming systems, and multi-agent AI applications. Seeking a role to apply my MLOps foundation and multi-cloud experience. I bring a tactical approach, positive "can-do" attitude, and great communication skills.

Education

University of California - San Diego, Major GPA: 3.72

La Jolla, CA

Bachelor of Science in Data Science - Halicioğlu Data Science Institute

Expected Graduation: Spring 2026

Technical Skills

Languages & Libraries: Python (pandas, NumPy), R, Java, C++, SQL, TypeScript, Scala, HTML/CSS, MATLAB

Tools & Frameworks: Matplotlib, Seaborn, TensorFlow, PyTorch, Scikit-Learn, CoreML, Apache Spark, Airflow, dbt, Kafka, Hadoop, Hive, NiFi, FAISS, Kubernetes, Docker, Git, Linux, Streamlit, Grafana, Tableau, Power BI

Cloud Platforms: Palantir Foundry + AIP, AWS (SageMaker, Glue, CodePipeline, S3, EC2), Snowflake, Azure Databricks

Certifications: AWS Cloud Practitioner (2023), AWS Data Engineer–Associate (2025), AWS Machine Learning (In Progress)

Experience

US State Department - US Embassy London

September 2023 – May 2024

AI/Data Science Developer Intern

- Built an AI document retrieval pipeline across 10,000+ policy docs using LLaMA 2 and FAISS vector search, achieving 95% retrieval accuracy and reducing policy Q&A response time from hours of manual search to under 10 seconds.
- Added rule-based filtering to embedding retrieval, cutting query latency by 60% and improving complex policy queries.
- Delivered 3 iterative prototypes over 8 months from NLP proof-of-concept to deploy-ready system in Agile sprints.

Projects (click 🔗 to view project)

PortOS – Border Traffic Orchestration AIP Agent 🔗 | *Palantir Foundry, AIP, TypeScript, Ontology*

2026

- Built an end-to-end AIP Agent on Palantir Foundry for U.S. Customs Port of Entry ops, designing a 5-object ontology data model with link schemas (Port, CrossingEvent, OfficerShift, Alert, NiiSystem) across 7 datasets.
- Developed 3 decision functions (TypeScript surge detector, lane-config recommender, AIP Logic wait-time predictor) driving 2 automations and real-time alerts in an operations dashboard.
- Modeled a reactive-to-predictive lane-staffing workflow projected to cut traveler wait times from 95 to 45 minutes (52% reduction) using CBP BWT and DHS OIG data.

Real-Time Big Data Pipeline on AWS 🔗 | *Kafka, PySpark, Hive, Apache Airflow, NiFi*

2025

- Built a batch-and-streaming ETL pipeline with Kafka (Avro), PySpark Structured Streaming, and Hive (Parquet) for distributed analytics storage.
- Orchestrated fault-tolerant workflows using Apache Airflow DAGs with automated task scheduling, managing 15+ dockerized services (Kafka, Spark, Hive, HDFS) for real-time data warehousing and trend analysis.
- Implemented data lineage with Apache NiFi for secure ingestion and encryption across distributed pipeline.

Gaussian Time Series MLOps Pipeline 🔗 | *AWS SageMaker, Terraform, CodePipeline, Flask, ECR*

2025

- Developed end-to-end MLOps pipeline on AWS using Gaussian Process Regression that achieved $R^2 = 1.0$ and MAE of 10.4 across 132 months of historical call center data, with fully automated CI/CD through CodePipeline.
- Implemented infrastructure-as-code with Terraform for reproducible ECR/Lightsail deployments and Flask API.
- Staged deployment with manual approval gates between staging and production environments.

Clinical Analytics with ML Drift Detection 🔗 | *Snowflake, Scikit-Learn, Power BI*

2025

- Developed Snowflake healthcare analytics processing 236K+ patient records with ML drift detection using 5 statistical tests (K-S, PSI, Chi-Square) and automated retraining at 10% performance degradation.
- Created Power BI dashboards for real-time model monitoring and clinical KPIs with automated retraining.
- Built alerting system for managing data quality and feature distribution shifts in production environments.

PhysioTwin – iOS Stress Detection App (Senior Capstone) 🔗 | *CoreML, SwiftUI, HealthKit*

Fall 25 – Winter 26

- Built a 3-stage CoreML pipeline distinguishing cognitive stress from physical exertion: Random Forest activity classifier (93.6% LOSO, 22 subjects), sleep-adjusted HR gating, and DC/AC PRSA stress score (0–100).
- Built an iOS app (HealthKit, CoreML, SwiftUI) that ingests real-time Apple Watch biometrics, with a continuous recovery score that drains with stress and recharges via logged recovery.
- Validated with 22-fold LOSO, ANOVA biomarker analysis, and 47 unit tests covering sleep, DC/AC, and HRV metrics.